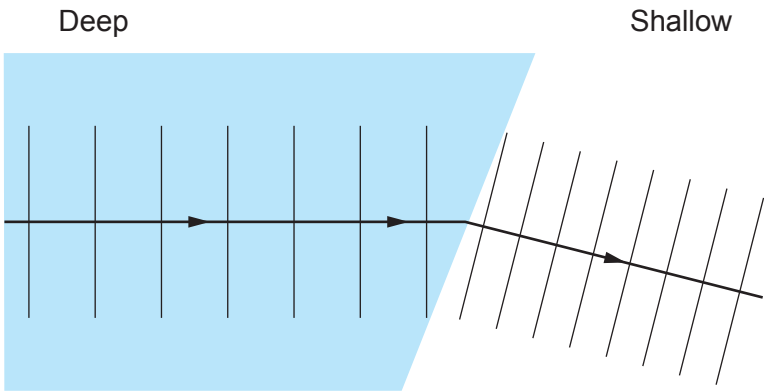


3. A ripple tank is used to demonstrate what happens to water waves as they travel from deep water into shallow water.

The results are shown in the diagram below.



(a) The water waves change direction at the boundary between deep and shallow water.  
State the name of this effect. .... [1]

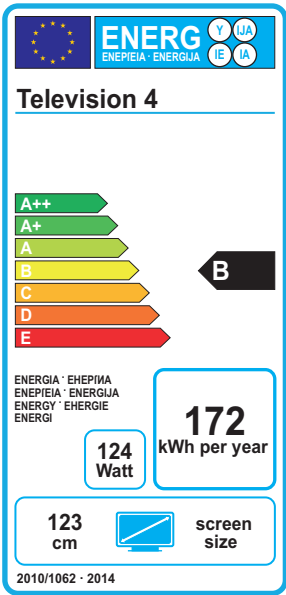
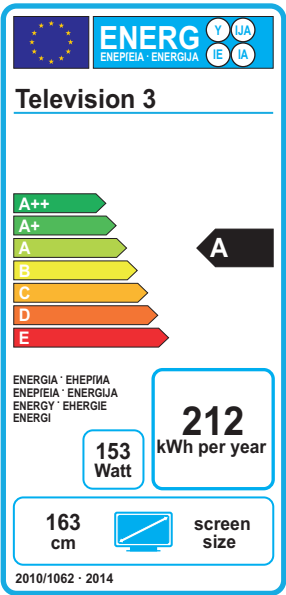
(b) Tick (✓) the boxes next to the **two** correct statements that describe the water waves. [2]

- The frequency of the waves is higher in shallow water ☐
- The amplitude of the waves is the same in deep and shallow water ☐
- The wave speed is less in shallow water ☐
- The wavelength decreases as the waves pass from deep to shallow water ☐

(c) In deep water the waves have a wavelength of 3 cm and a frequency of 5 Hz. Use this information and an equation from page 2 to calculate the wave speed. [2]

Wave speed = ..... cm/s





Information on the cost of each television is given below.

| Television | Purchase cost (£) |
|------------|-------------------|
| 1          | 280               |
| 2          | 1 000             |
| 3          | 1 500             |
| 4          | 800               |

(a) Use the information from the labels and the table to tick (✓) the **three** correct statements below. [3]

- Television 1 uses less energy per second than television 2
- The largest televisions always use the most energy
- The purchase cost of television 2 is 1.5 times that of television 3
- More expensive televisions always use less energy
- Television 3 uses 40 units more per year than television 4
- Televisions with the same energy rating, e.g A+, don't always have the same power
- ☐
- ☐
- ☐
- ☐
- ☐
- ☐



- (b) It is claimed that power is proportional to screen size. Use the data for television **1** and television **2** to determine if this claim is true. [3]  
*Space for calculations.*

.....

.....

- (c) (i) Use the equation:

$$\text{time (h)} = \frac{\text{units used (kWh)}}{\text{power (kW)}}$$

and data from the energy label to calculate how many hours the label suggests that television **2** is used for in 1 year. [2]

Hours used = .....

- (ii) Use an equation from page 2 to calculate the cost of using television **2** for 1 year if 1 unit (kWh) of electricity costs 16 p. Give your answer in pounds (£). [2]

Cost = £ .....



Examiner  
only

- (iii) The expected lifespan of a television is 10 years. Simon concludes that it will be more cost effective to buy and run television **2** for 10 years but Sarah disagrees and claims that television **4** will be cheaper. Use the data to determine who is right. [3]  
*Space for calculations.*

.....  
.....

- (iv) Other than to save money, why should consumers be encouraged to choose appliances that use less energy? [2]

.....  
.....

**END OF PAPER**

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